Overview of Antennas and Their Properties

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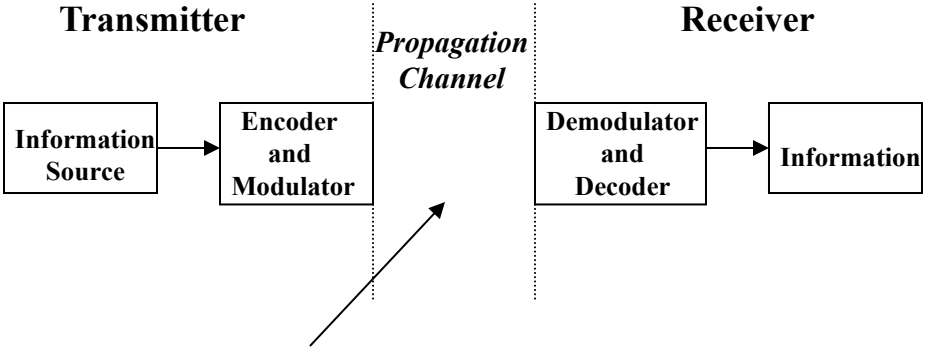
Lecture Objective

By the end of this lecture, you will be able to:

- Understand the basic functionality of an antenna and its role in a system.

Basic Communications System



The "connection" between the transmitter and the receiver: coaxial or fiber optical cable; or antennas and some path between them

What is an Antenna?

- Webster's dictionary: "a usually metallic device (as a rod or wire) for radiating or receiving radio waves."
- *The IEEE Standard Definitions of Terms for Antennas*, IEEE Std 145-1983: "a means for radiating or receiving radio waves."

An Antenna is a Transducer

- An antenna is a “transducer” that converts electrical signals (guided waves) into electromagnetic radiation (spherical waves) propagating in free space.
- A ***transducer*** is a device that converts one form of energy to another.
- Another example of a transducer that we are all familiar with is an audio speaker – which converts electrical impulses to sound waves.

Basic Operation of Transmit and Receive Antennas

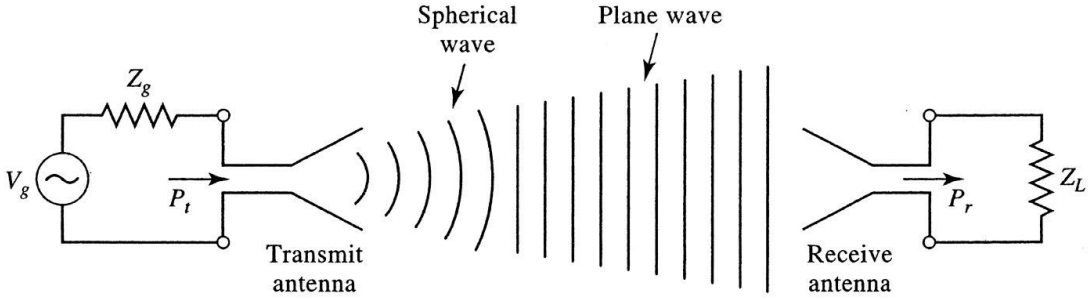


Figure 14.2
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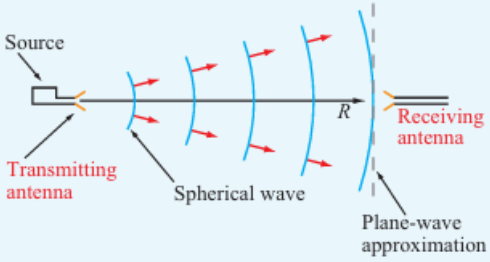


Figure 9-3 Far-field plane-wave approximation.

Some Characteristics of Antennas

- For satisfactory performance the antenna must usually be a significant fraction of a wavelength in size.
- Antennas that are less than about 0.1 wavelengths in all dimensions are called “electrically small” antennas and present significant challenges to their use in systems.
- A rigorous mathematical description of antenna radiation is given by ***Maxwell’s equations***.
- The behavior of an antenna is determined by the ratio of its size to the wavelength at which it is operating.

Some Types of Antennas

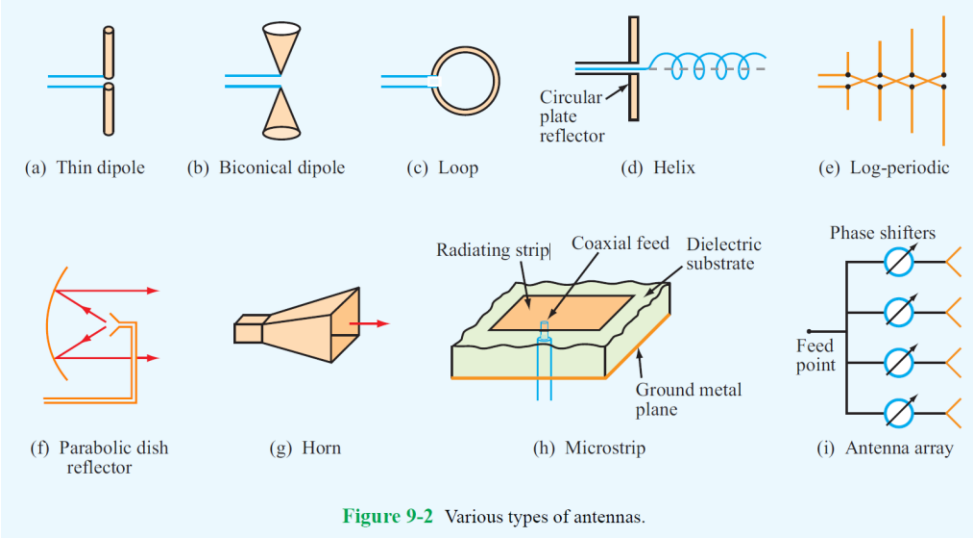


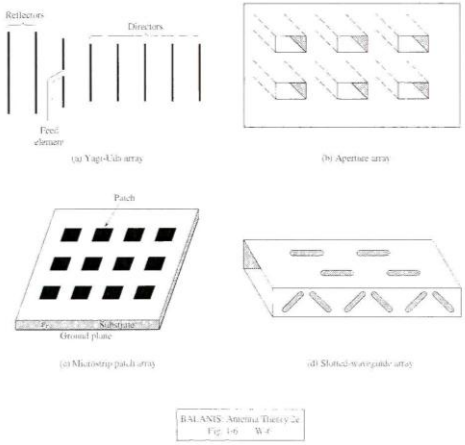
Figure 9-2 Various types of antennas.

Two Major Categories of Radiation Sources

- Current sources
  - Wire antennas
  - Microstrip antennas
- Aperture sources
  - Open waveguide
  - Waveguide slot array
  - Horn antenna

Array Antennas

- A geometric and electrical arrangement of (often identical) radiating elements
- Used to achieve radiation properties that cannot be obtained using a single element
- By controlling relative amplitudes and phases of each element's excitation, it is possible to realize a phased array with control over the shape of the radiation pattern and the direction of maximum radiation.



Antenna Theory: Analysis and Design, 2nd Edition, Constantine A. Balanis

Radiation Pattern (1 of 2)

- An antenna ***radiation pattern*** is described by a "mathematical function or a graphical representation of the radiation properties of the antenna as a function of space coordinates."
- Radiation properties include ***radiation intensity***, field strength, ***directivity***, ***axial ratio***, etc.

Radiation Pattern (2 of 2)

- Most often the term *radiation pattern* refers to a graphical representation of the radiated power at constant distance (r) in the *far-field region* as a function of direction (θ and ϕ).
- Often the pattern is plotted versus θ (ϕ) for a relatively few values of ϕ (θ).

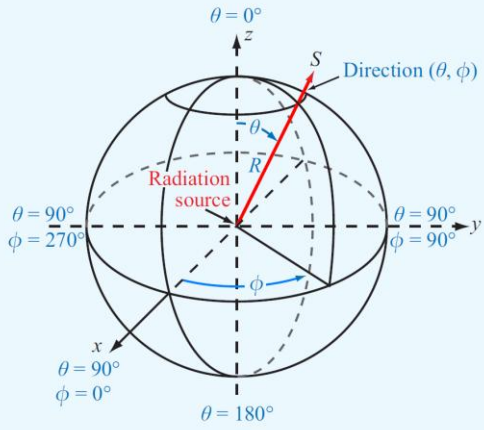


Figure 9-5 Spherical coordinate system.

Isotropic, Directional, and Omnidirectional Patterns

- An *isotropic radiator* is “a hypothetical lossless antenna having equal radiation in all directions.”
- A *directional antenna* has “the property of radiating or receiving electromagnetic waves more effectively in some directions than in others.”
- An *omnidirectional* antenna has an azimuthally symmetric radiation pattern.

Questions

The radiation power density from an isotropic antenna at a distance from its phase center of 100 meters is 1 mW/m^2.

1. What is the total power radiated by the antenna?

2. What is the radiation power density at a distance of 10 km from the phase center of the antenna?

Reciprocity

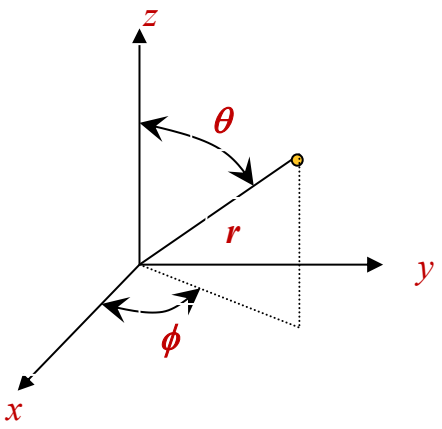
- Most antennas are **reciprocal** devices, exhibiting the same radiation characteristics for transmission and reception.
- Reciprocity makes antenna analysis very convenient. For example, we can compute the radiation properties of the antenna when it is transmitting and use this information even when the antenna is used to receive signals.

Azimuth and Elevation (1 of 2)

- Sometimes the directional characteristics of an antenna are specified in terms of *azimuth* and *elevation*.
- *Azimuth* refers to the angle with respect to a fixed line in the plane of the ground.
- *Elevation* refers to the angle with respect to the plane of the ground.
- The relationship between azimuth and elevation and θ and ϕ depends on how we set up the coordinate system.
- Azimuth and elevation planes are often called the *principal planes* of the spherical coordinate system used to describe the radiation pattern of the antenna.

Azimuth and Elevation (2 of 2)

- Assume the coordinate system is set up such that the +z-axis is pointing to the zenith.
- θ is the *zenith angle*.
- *The azimuth plane* refers to the θ = *90°* plane.
- *The elevation plane* to any ϕ = *constant* half-plane.
- The elevation angle is *90°*- θ



Field Regions

- The behavior of the field generated by an antenna depends on the distance from the antenna.
- We define three regions within which the radiated field of the antenna exhibits different features:
  - ***Reactive near field region***
  - ***Radiating near-field region (Fresnel region)***
  - ***Far-field region (Fraunhofer region)***

Far-Field (Fraunhofer) Region (1 of 2)

- The *far-field (Fraunhofer) region* is "that region of the field of an antenna where the angular field distribution is essentially independent of the distance from the antenna."
- The radiated fields are *TEM to r*, i.e. locally the field behaves like a plane wave.

Far-Field (Fraunhofer) Region (2 of 2)

- Usual far-field criterion:

$$R > \frac{2D^2}{\lambda}$$

D = maximum dimension of antenna
R = distance from the observation point to the ***phase center*** of the antenna.

The point from which the spherical waves appear to emanate

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